

Ethics & Fairness Memo: Credit Card Fraud Detection Model

Project: Credit Card Fraud Detection ML Pipeline

Model: Random Forest Classifier with SMOTE Resampling

Dataset: 100,000 transactions (1% fraud rate)

Performance: 61.7% fraud detection, 32.55% false alarm rate

Status:  Not Ready for Production

1. Dataset Representativeness

The current model was trained on a highly imbalanced dataset (99% normal, 1% fraud). While techniques like SMOTE were used to technically balance the training data, synthetic data does not introduce new real-world patterns.

Furthermore, public datasets from platforms like Kaggle often lack transparency regarding the demographic and geographic distribution of the cardholders. If the data originates from a specific demographic (e.g., predominantly middle-to-upper-class users in a single country), the model will not accurately represent or generalize to low-income users or international transactions.

2. Ethical Risks

The current model operates with a 32.55% false alarm rate, meaning roughly 1 in 3 normal transactions are flagged as fraudulent.

Disparate Impact

The burden of these false alarms is unlikely to be distributed equally. Location_encoded is the single most important feature (5.12% importance). If the model over-indexes on location as a proxy for

fraud, marginalized communities will experience account freezes at much higher rates than affluent communities. This creates digital redlining and disparate impact on vulnerable populations.

Lack of Explainability

Because Random Forest is an ensemble model, it operates as a "black box." When a customer's card is declined, the institution cannot easily provide a clear, exact reason why that specific transaction was blocked, violating principles of transparency and fairness.

High Precision Cost

With a precision of only 0.02 (2%), 98 out of 100 transactions flagged as fraud are actually normal. This unacceptable false positive burden makes autonomous decision-making unethical.

3. When the Model Should NOT Be Used

Fully Automated Account Closures: This model should absolutely not be used to automatically close accounts or permanently ban users. With a precision of only 0.02, 98% of the transactions it flags are actually normal.

Without Human-in-the-Loop: It should only be used as an alert system to trigger SMS verification ("Did you make this purchase?") or route cases to human fraud analysts for manual review.

In New Geographic Markets: The model should not be deployed in regions not represented in the training data, as the Location_encoded feature will fail or misclassify transactions from unfamiliar geographies.

4. Potential Real-World Harm

Financial Paralysis During Emergencies

A false positive (card decline) could prevent a user from paying for urgent medical care, emergency travel, or essential services like

towing. A 32.55% false alarm rate means this happens to millions of customers annually.

Erosion of Trust

A 32.55% false alarm rate will cause "alert fatigue" for both customers and internal fraud analysts. Customers become frustrated and may abandon the financial institution entirely. Analysts ignore legitimate fraud alerts due to excessive noise.

Stigmatization

Repeatedly declining a user's card at the point of sale causes public embarrassment and psychological distress. This harm is particularly severe if users perceive they are being targeted based on where they live or what stores they shop at, reducing financial inclusion for vulnerable populations.